Lahars

What are they, what can they do, and what do we need to know to model them?





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What is a lahar?

- Rapid, concentrated flows of rock, mud, and water from volcanoes
- A high-concentration “flash flood” from a volcano
- Up to 2/3 solids, 1/3 water
- Can flow up to 40 km/h
- Can flow >200 km
- **Destructive**



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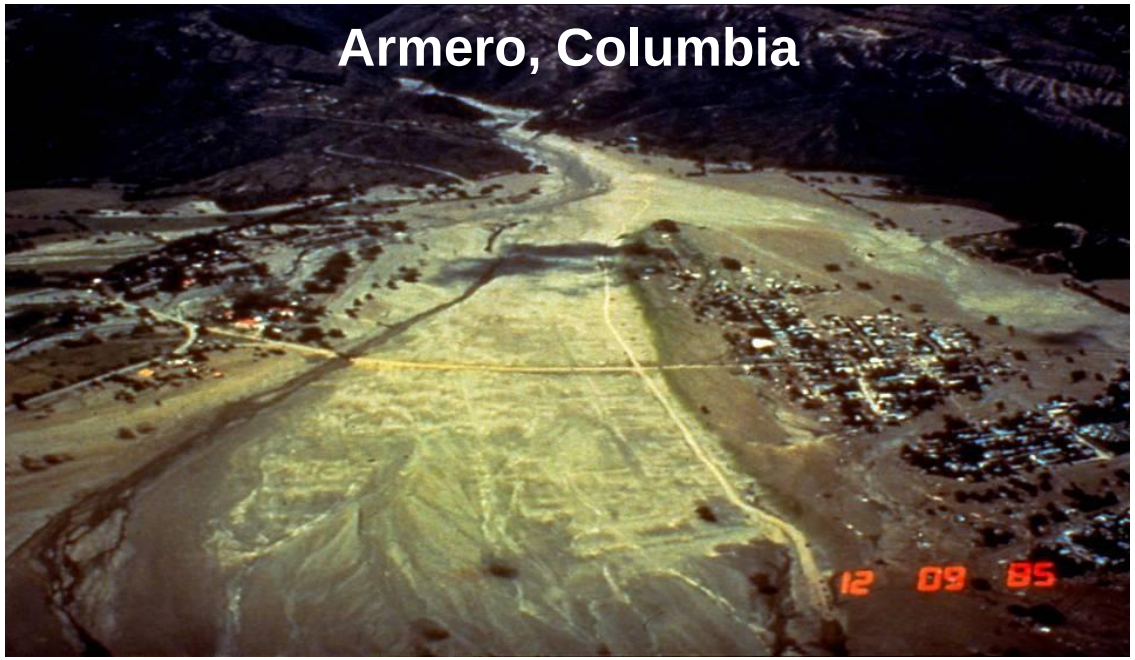
Lahars can occur

WITH or WITHOUT eruptions!!

Triggered by:

- Rapidly melting snow/ice (from dome growth, PDCs, incandescent ejecta, lava flows)
- Heavy rain on tephra or other sediment (meteoric or eruption-induced)
- Lake breakouts (eruption-related or not)
- Sediment dam breakouts (eruption-related or not)
- Glacier outburst floods (jökulhlaups caused by eruptions under glaciers)
- Gravitational collapse of edifice (landslides)
- Groundwater expulsion



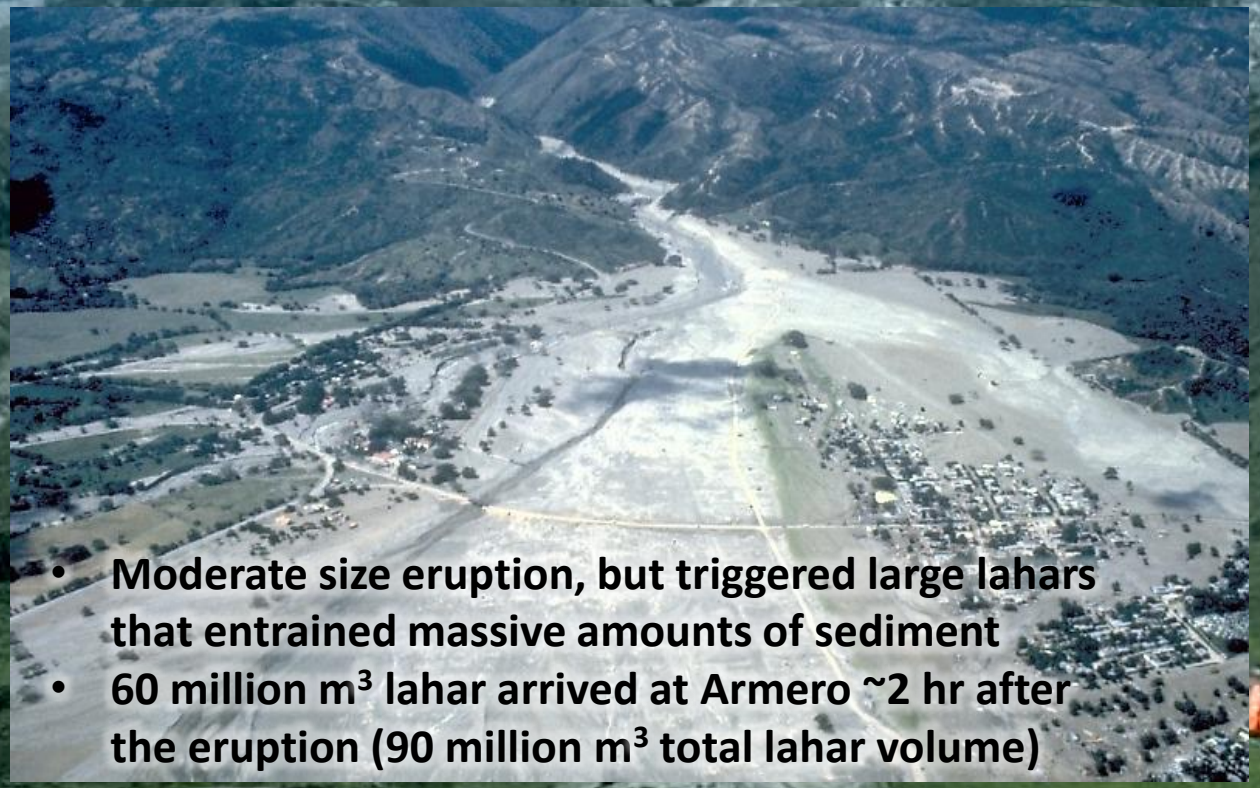




NEVADO DEL RUIZ

Altitude:	~5,390 m (17,680 ft)
Ice cap:	~17 km^2
Eruption:	Nov. 13, 1985, VEI 3
Meltwater produced:	~2 x 10^7 m^3
Total lahar volume:	~ 8 x 10^7 m^3
Lahars reached:	> 100 km (3 valleys)
Fatalities:	> 23,000 people
Previous lahars:	1595 and 1845

Ruiz Volcano & Armero, Columbia, The 1985 tragedy that led to VDAP



- Moderate size eruption, but triggered large lahars that entrained massive amounts of sediment
- 60 million m^3 lahar arrived at Armero ~2 hr after the eruption (90 million m^3 total lahar volume)

How big, how far, how frequent?

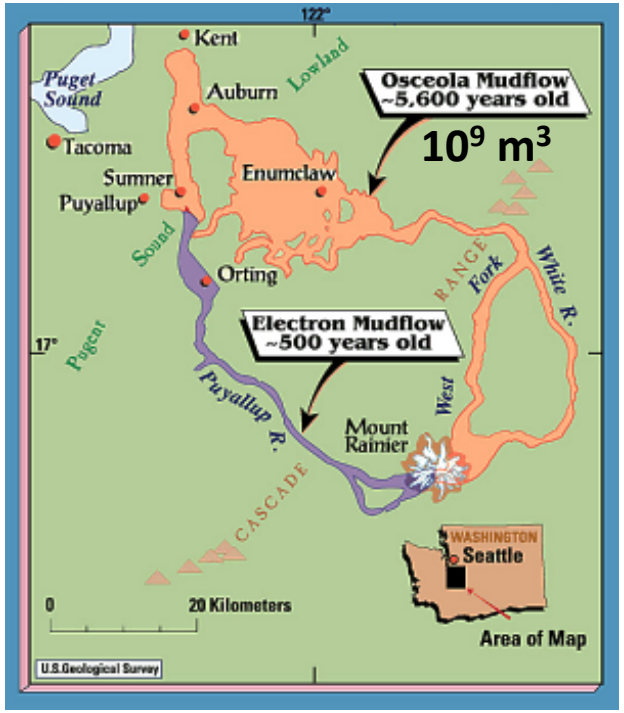
- Small: Up to 10^4 m^3, up to 10 km, once or more per decade
- Large: 10^4 to 10^6 m^3, up to 50 km, once or more per century
- Very large: 10^6 to 10^8 m^3, up to 100 km, once per 100 to 1,000 years
- Extremely large: 10^8 to 10^{10} m^3, >200 km, once per 1,000 to 10,000 years



Polallie Creek, Mount Hood, 1980

(10^3 - 10^5 m^3)





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Flow Types

Stream Flow

Hyperconcentrated Flow

Debris Flow & Lahar

Granular Flow









wet **dry**

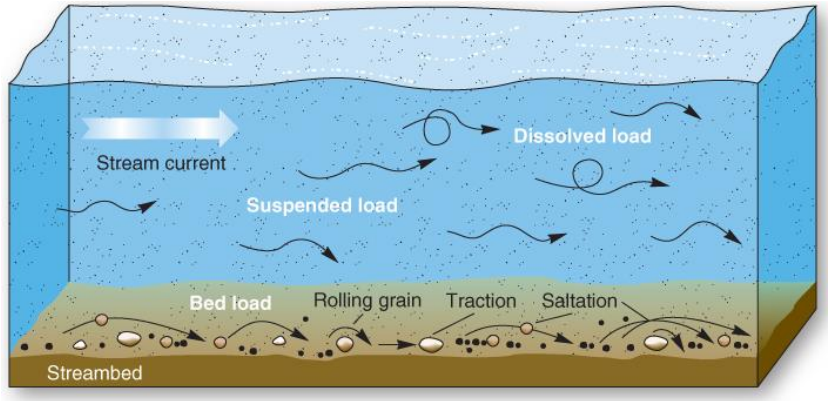
Volume fraction between 100% fluid flow --- 100% granular flow

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Flow Types



Streamflow:
Water flow with low suspended sediment load; transport of grains during high flow events by suspension, saltation & bedload dragging; no shear strength; good fishing!



▲Figure 15.13 **Fluvial transport.** Fluvial transportation of eroded materials as dissolved load, suspended load, and bed load. Bed load includes grains transported by traction (rolling or dragging of materials along the streambed) and saltation.

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Hyperconcentrated Flow:
Muddy flow with high suspended sediment load; transport of grains during high flow events by suspension, saltation & bedload dragging;
no matrix strength, not a cohesive flow; erosion and undermining of structures can be very damaging as can the extreme aggradation result in massive, damaging sediment deposits.





Flow Types



Debris Flow and Lahar:
Slurry flow of concrete-like mud and rock with high cohesion where the process is driven by intergranular pressures (pore fluid pressure); has a yield strength to buoyantly transport larger clasts, boulders, trees, bridges; great destruction and loss of life is possible

Granular Flow:
Granular flow or rock avalanche is dominated by frictional flow and collisions



Stream Flow

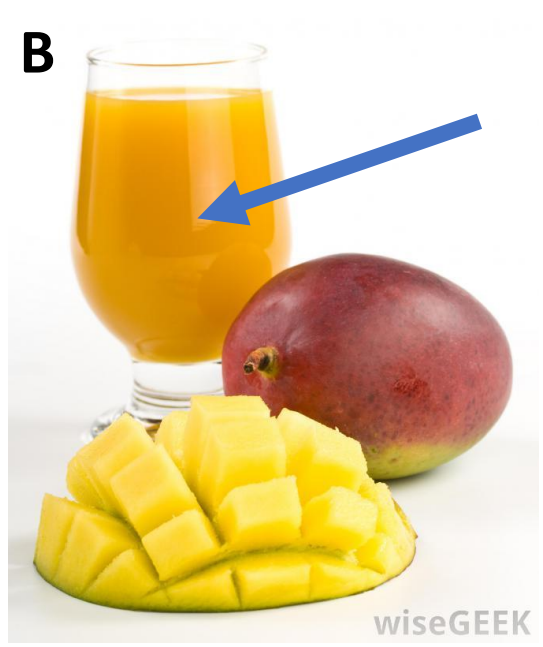
Hyperconcentrated Flow

Debris Flow/Lahar

Granular Flow

Flow Type









VolFilm

https://vimeo.com/volfilm/albums

Impacts:
https://vimeo.com/208038678

Experience:
https://vimeo.com/258992598

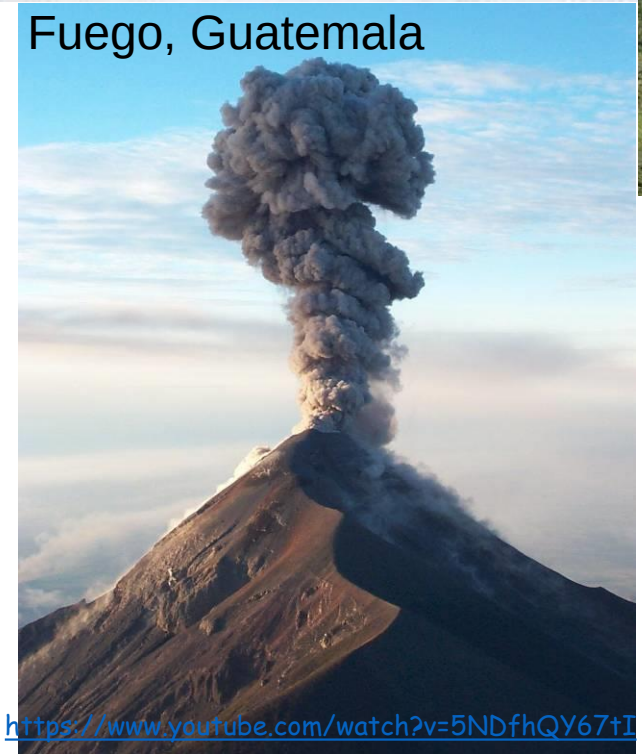
Lahar Videos







https://www.youtube.com/watch?v=aauPb-gQLNA





Takahashi, 1981

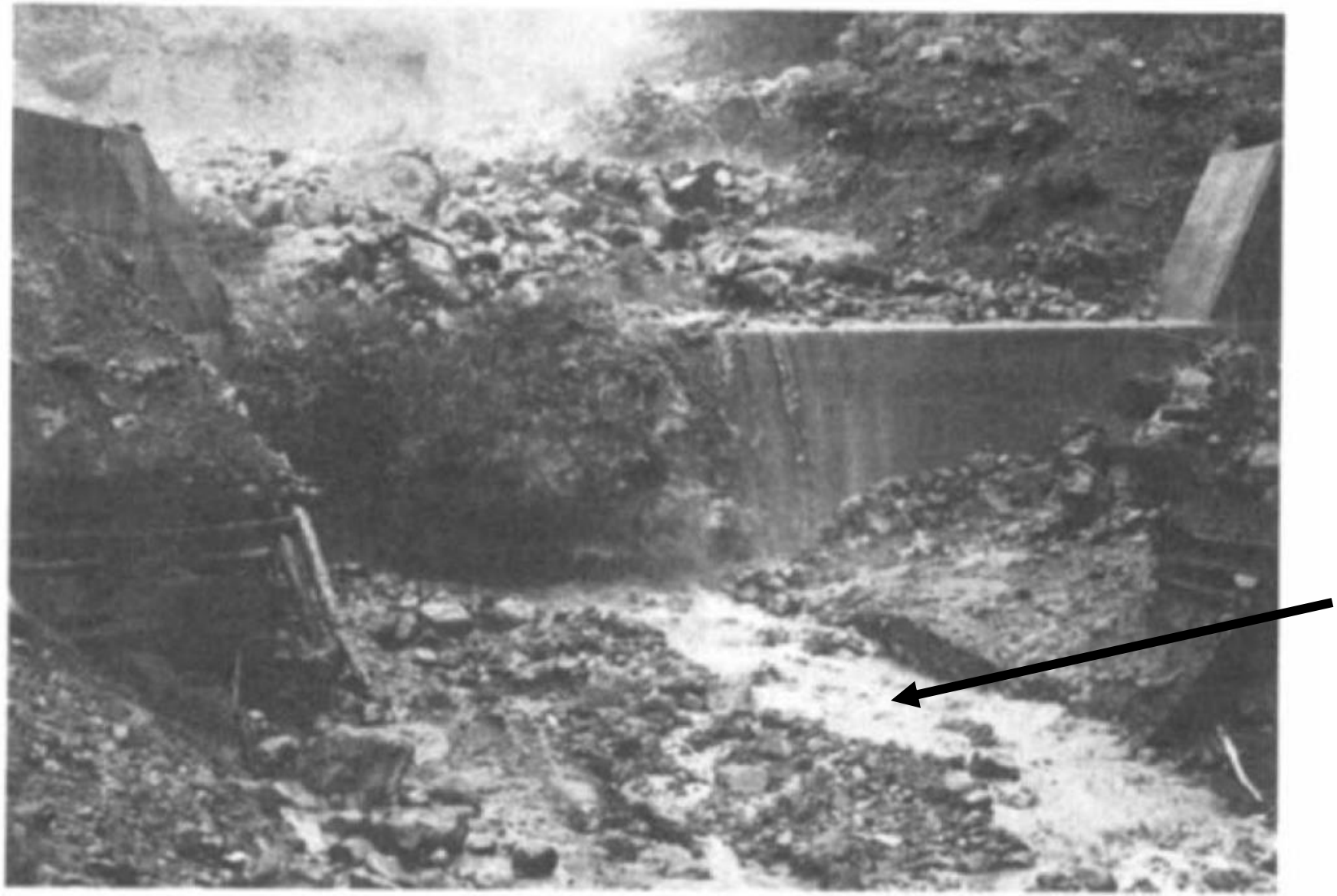


Photo 1. The front is coming. Note the scanty discharge in advance of the debris flow.

Takahashi, 1981

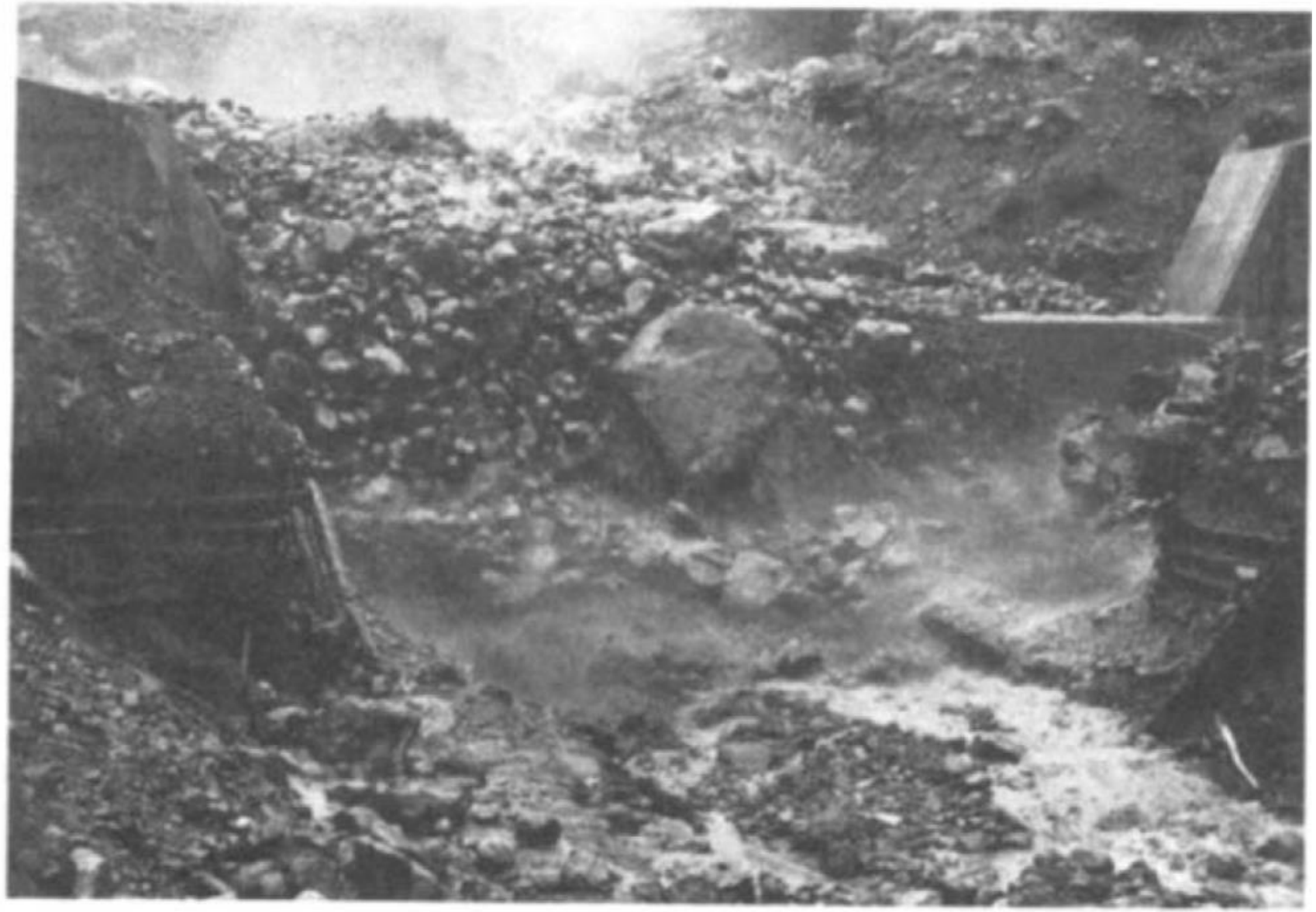


Photo 2. 2 seconds later than Photo 1. The big stone is 2.5 m in diameter.

Takahashi, 1981



Photo 3.

Takahashi, 1981



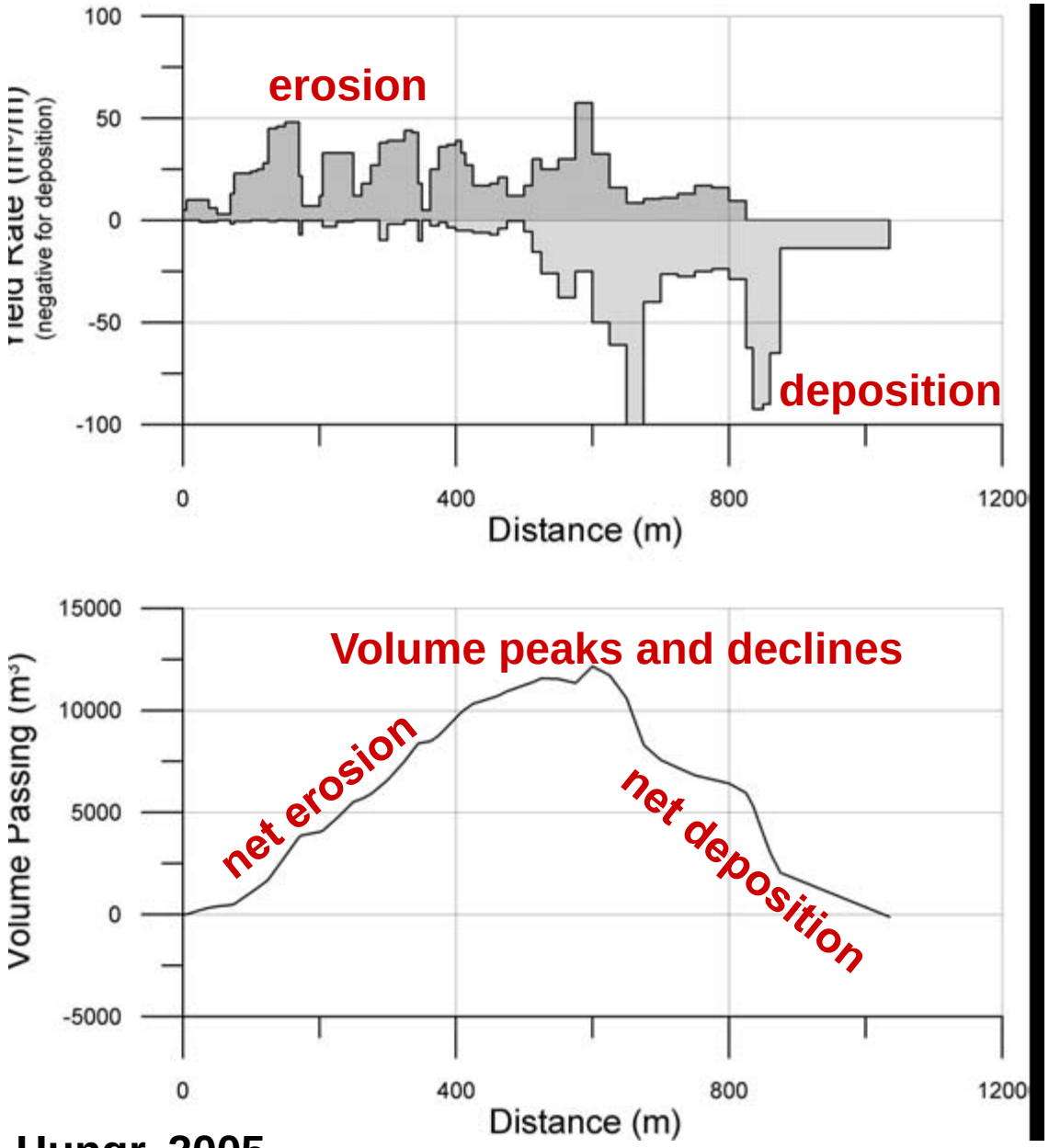
Photo 4.

Mass Balance Analysis

Yield Rate ($\Delta m^3/m$)

Change in volume per meter distance along channel

Flow Volume (m^3)



Hungr, 2005

- Not constant as flow moves from beginning to end; increases and decreases and it varies.
- Maximum volume along the path marks an important change in lahar process; erosion rate slows and deposition rate increases.
- Erosion is greatest on steep slopes where there is much loose volcanic material
- Deposition is greatest on gentle slopes

Lahar Deposits
(different from the lahar itself)
Mount St. Helens

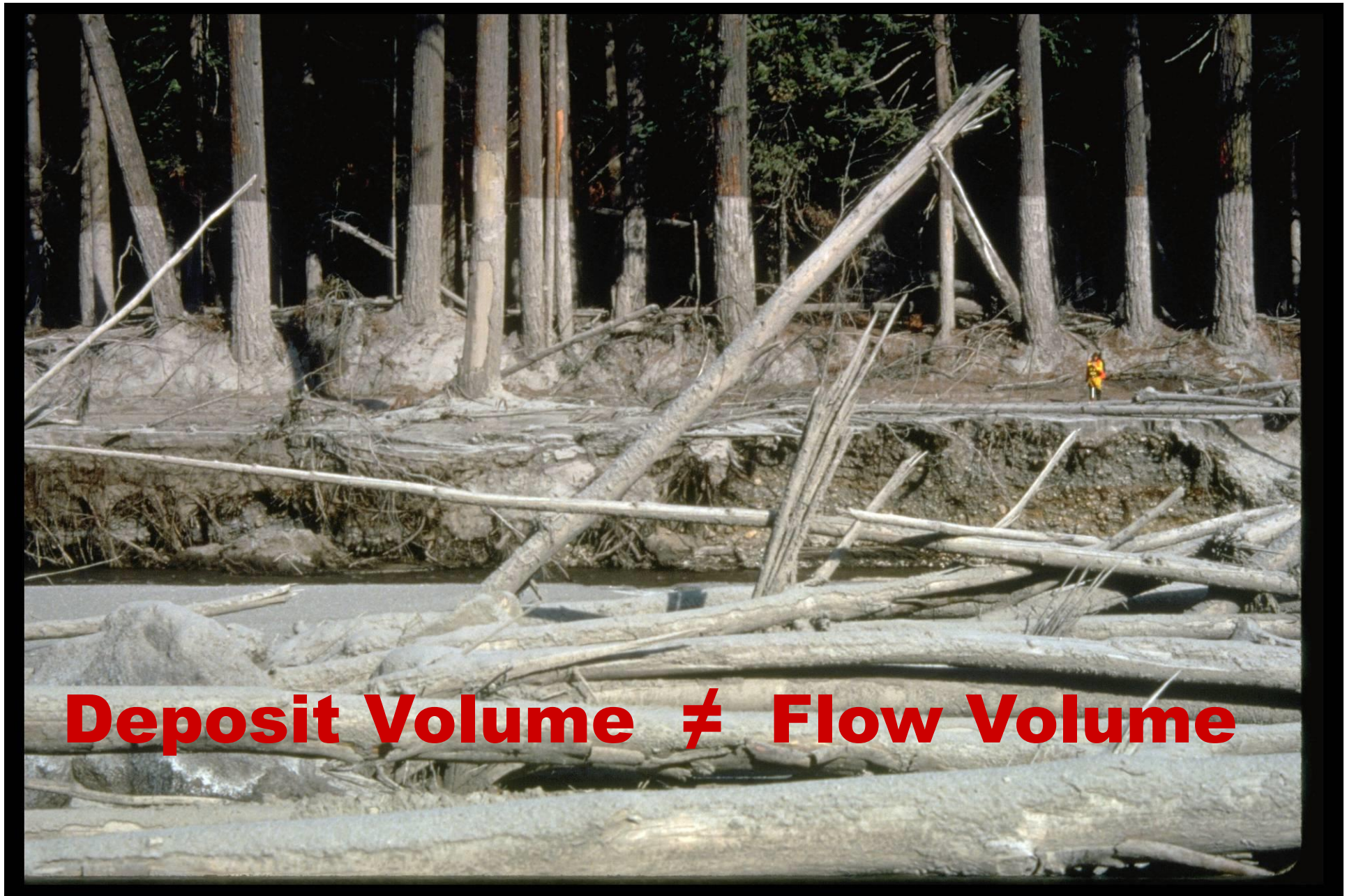


Deposit Volume < Flow Volume

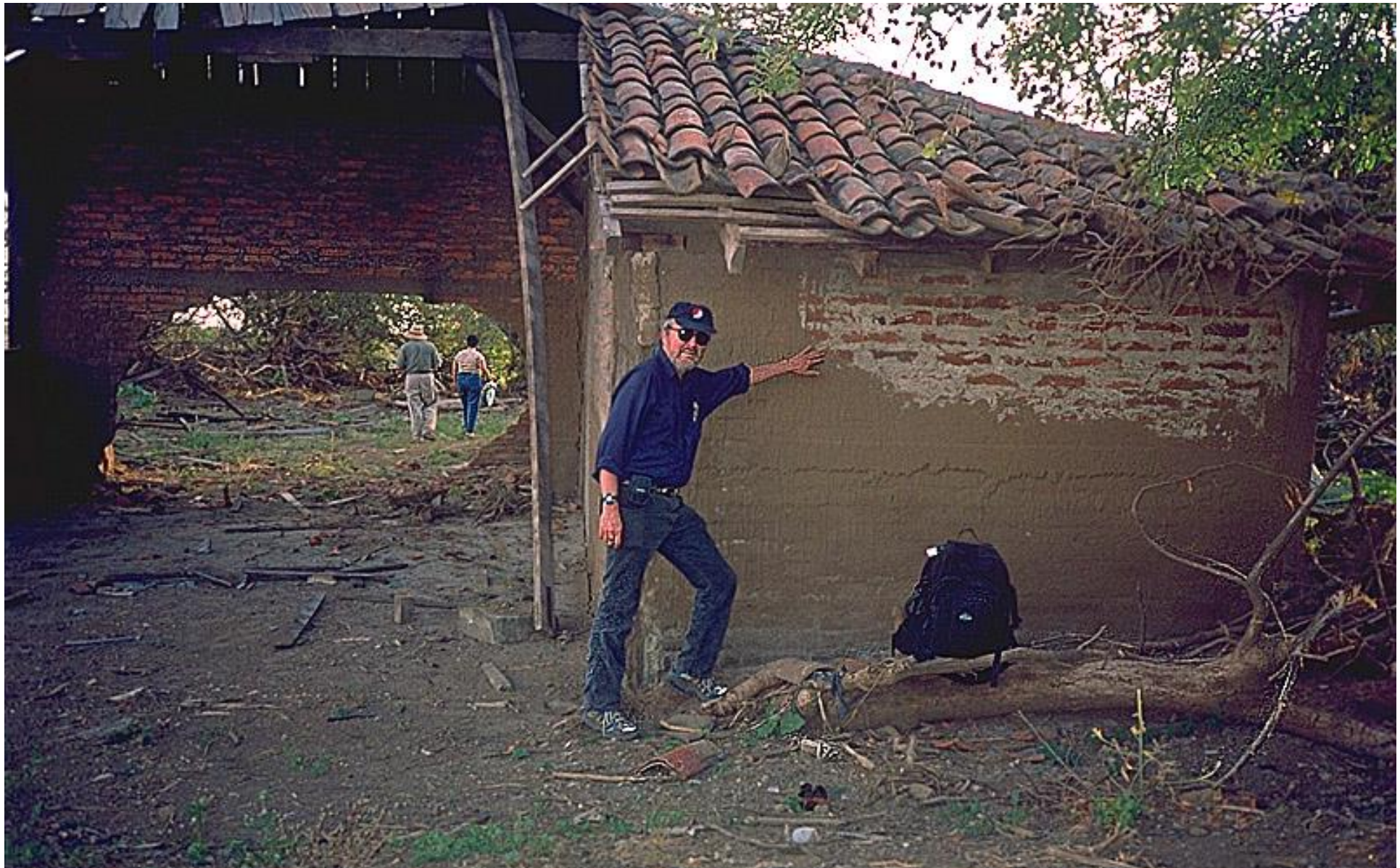
Lahar Deposits- Mount St. Helens



Lahar Deposits- Mount St. Helens



Casita Volcano, Nicaragua:
Mud line on the last remaining structure in El Porvenir shows height of the lahar--all other homes were completely destroyed and carried away. At this distance from the volcano, the lahar was nearly 1.5 km wide with an average depth of 3-4 m.



Lahar Deposits- Mount St. Helens



After a Lahar- Pinatubo



After a Lahar- Pinatubo



After a Lahar- Pinatubo

Hyperconcentrated flow

Sediment drains



After a Lahar- Concepcion

Hyperconcentrated Flow

Channel avulsion

Flooding



Channel Changes with Time– Relevant to DEM vintage

Rio Paez lahars at Belalcazar, Colombia; 1994, 2007, 2008





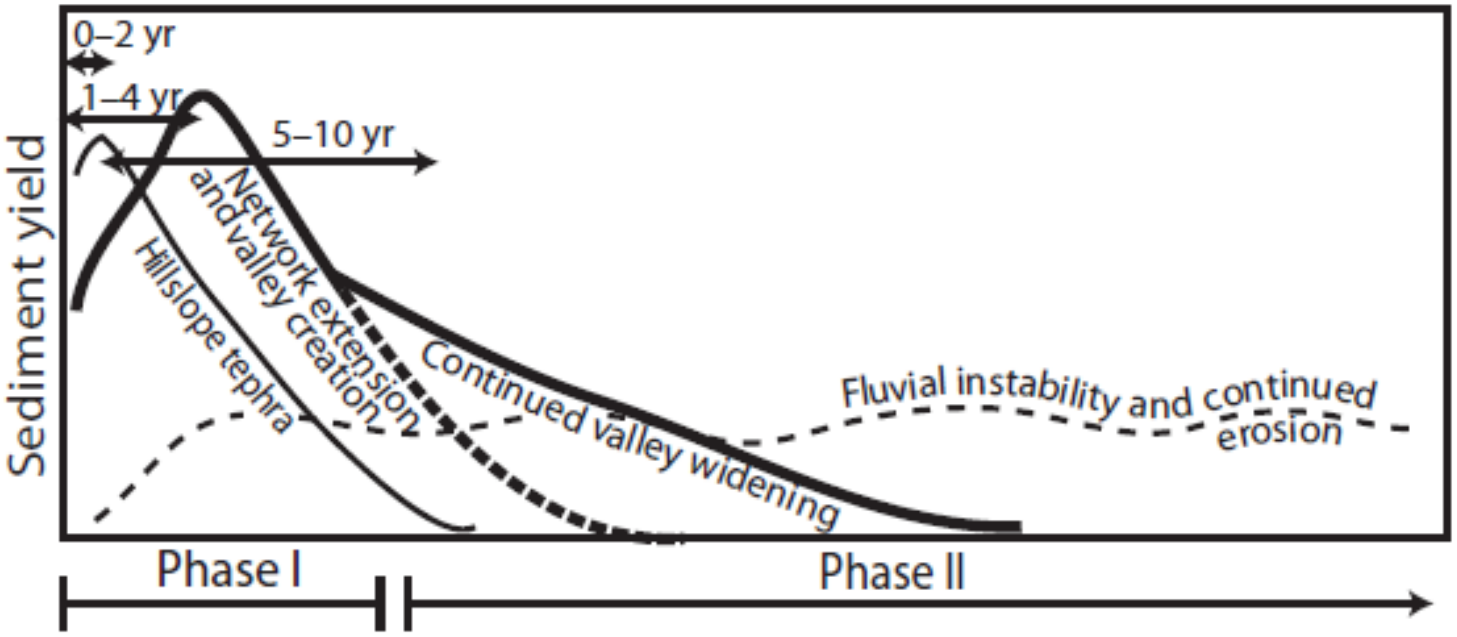




After events, channel may change course or may become choked with sediment that will result in subsequently higher and wider flooding zones; also after some quiescence, dramatic incision may also occur. Updating DEMs will be needed for future hazard maps.

Sediment yields

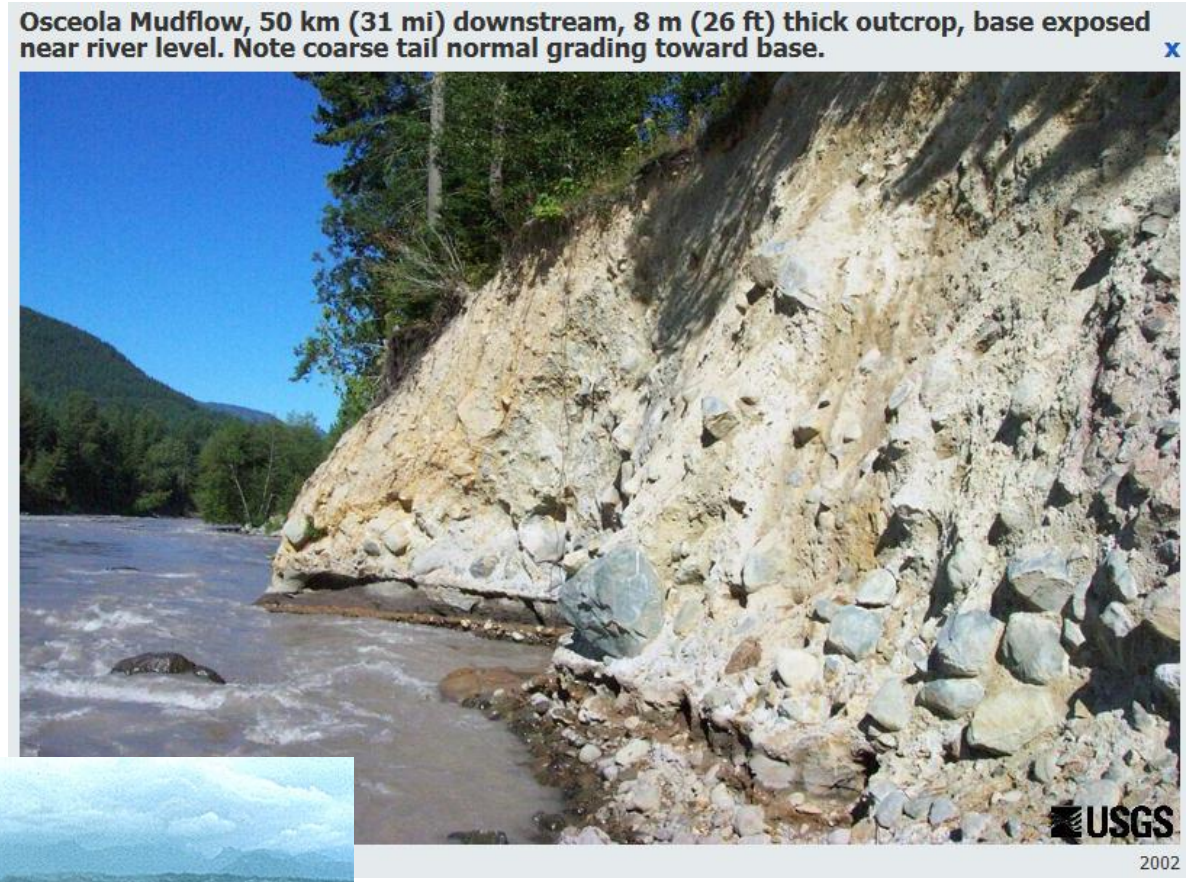
- Work at Pinatubo and Mt. St. Helens has shown that sediment yields can remain elevated for decades
- Can increase flood peaks and provide source material for lahars
- Schematic illustrating change of dominant sediment sources through time.
- Phase I: hillslope tephra and valley creation lead to high sediment yields early that decline exponentially through time.
- Phase II: continued valley widening and fluvial instability maintain high sediment yields for decades.



Gran et al. 2011

Studying lahars

1. Field observation and mapping older deposits at Mount Rainier, Washington, USA





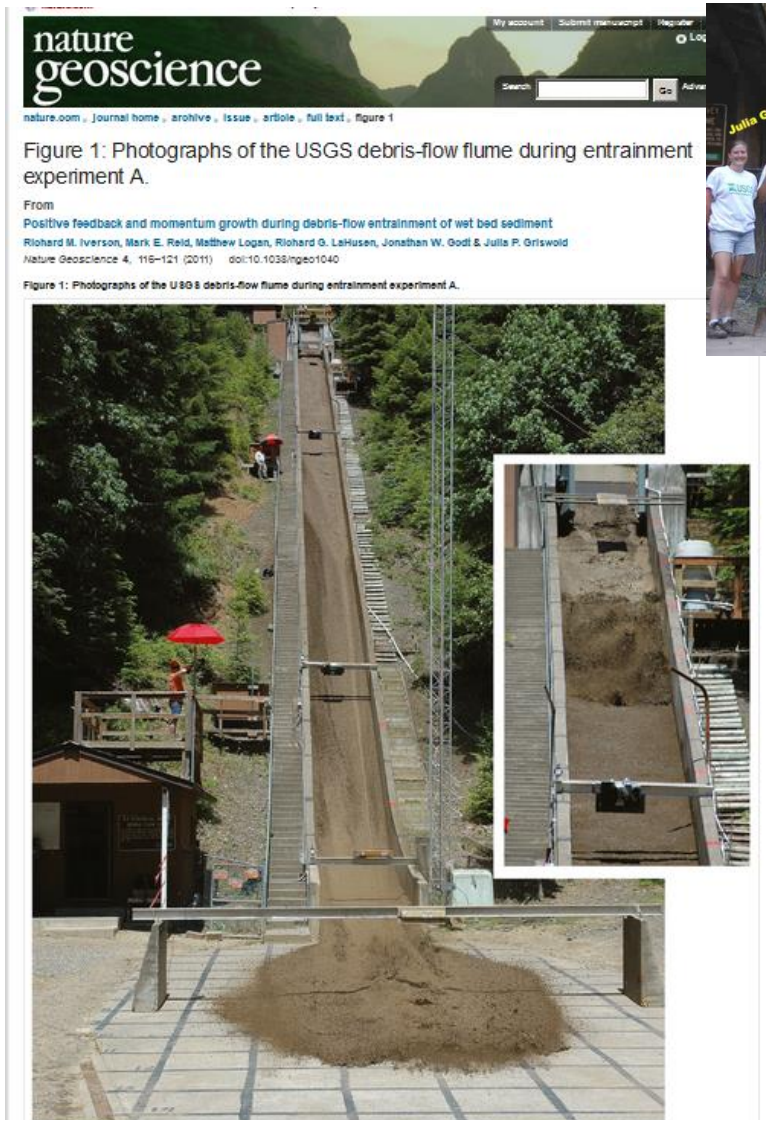
2. Experience of lahars and their consequences

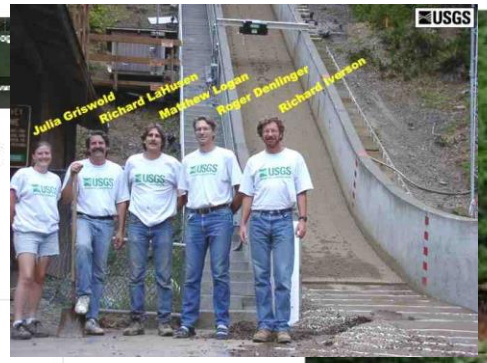
- Nevado del Ruiz; location of the former town of Armero, Colombia
- Pinatubo, Philippines

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3. Experiments to demonstrate field observations; research; confirm modeling; prediction.

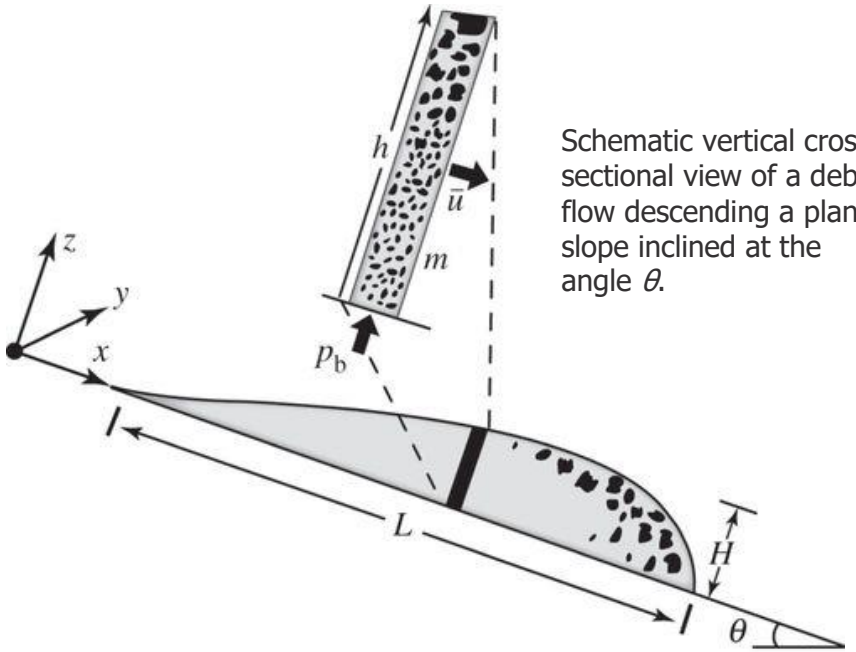
>163 debris flow experiments 1992-2017
>18 hours edited video online
http://pubs.usgs.gov/of/2007/1315/



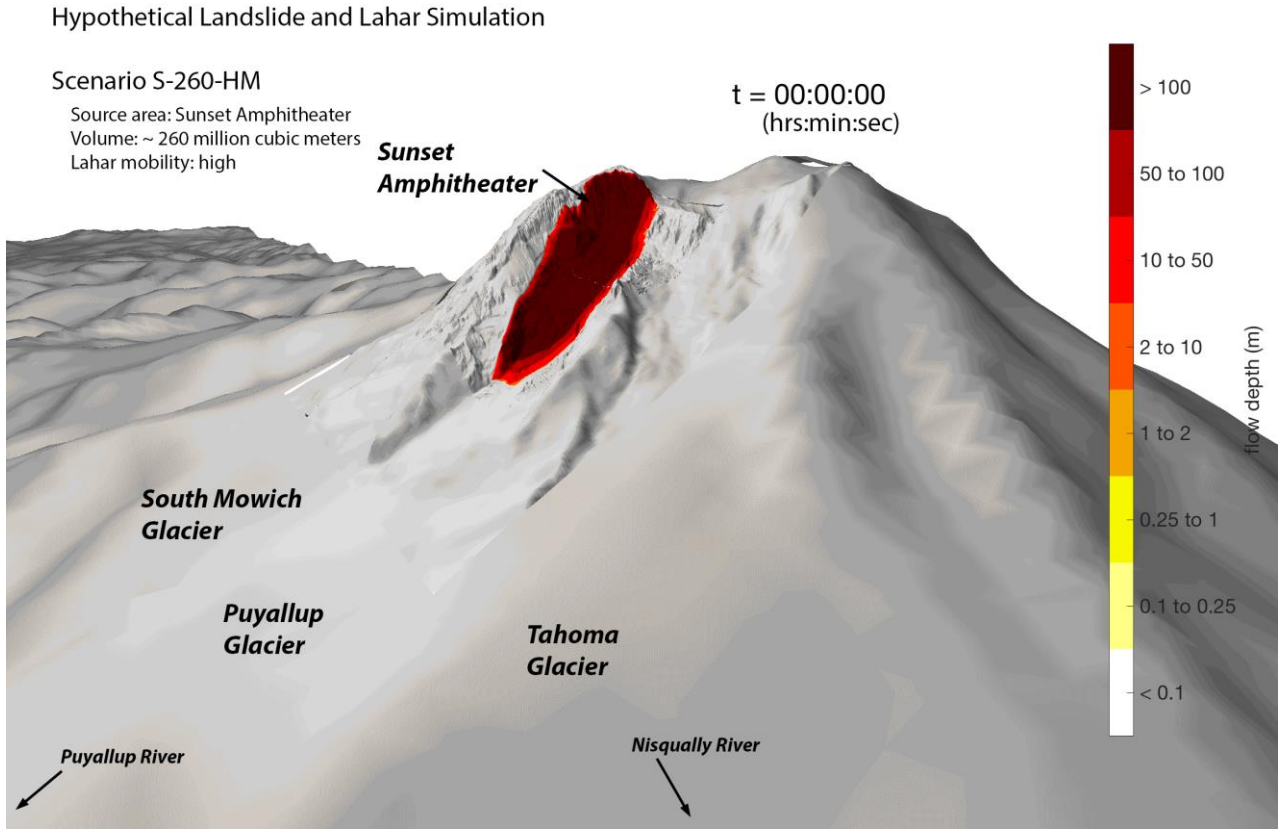




4. Mathematical modeling to understand the dynamics and to make predictions (D-CLAW)



Schematic vertical cross-sectional view of a debris flow descending a planar slope inclined at the angle θ.



George et al. 2021 https://pubs.usgs.gov/publication/ofr20211118

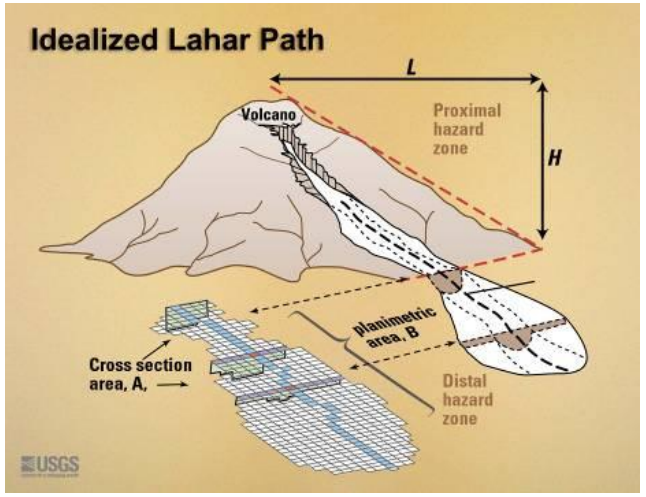
David George, Mary Benage, Richard Iverson, Roger Denlinger



George & Iverson 2014: https://doi.org/10.1098/rspa.2013.0820
Iverson & George 2014: https://doi.org/10.1098/rspa.2013.0819

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5. Empirical and Statistical modeling to understand the limits of potential, average inundation and to help make general predictions (LAHARZ)



Dick Iverson, Steve Schilling, Jim Vallance, Julie Griswold



Iverson & Schilling 1998: https://doi.org/10.1130/0016-7606(1998)110%3C0972:ODOLIH%3E2.3.CO;2
Schilling 1998: https://doi.org/10.3133/ofr98638
Griswold & Iverson 2007: https://pubs.usgs.gov/sir/2007/5276/
Schilling 2014: https://pubs.usgs.gov/of/2014/1073/



LaharZ model (left) of the possible lahar hazards from the Carbon and Puyallup River drainages from Mount Rainier. The town of Orting, Washington (photograph) sits in the lahar hazard zone. https://www.usgs.gov/media/images/laharz-model-possible-lahar-hazards-carbon-and-puya

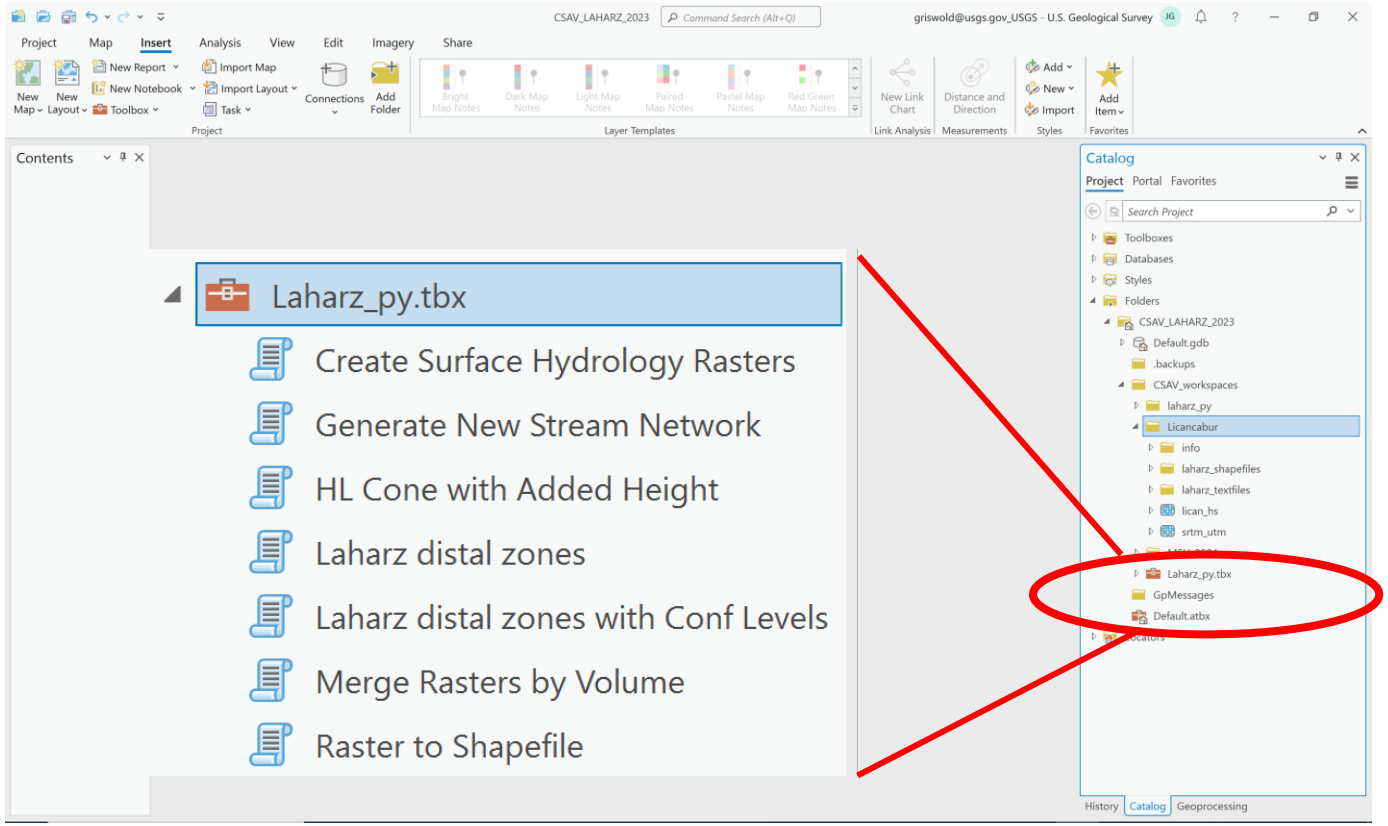
CSAV 2024 & beyond: LAHARZ has options!

- ArcGIS ArcMap 10.x
- Laharz_py ArcGIS toolbox
- Schilling et al. 2014
- https://pubs.usgs.gov/of/2014/1073/
- CSAV file folders



CSAV 2024 & beyond: LAHARZ has options!

- ArcGIS Pro
- Julie Griswold 2023 rewrite of Schilling et al. 2014 code for ArcGIS Pro
- Email me!
- CSAV file folders

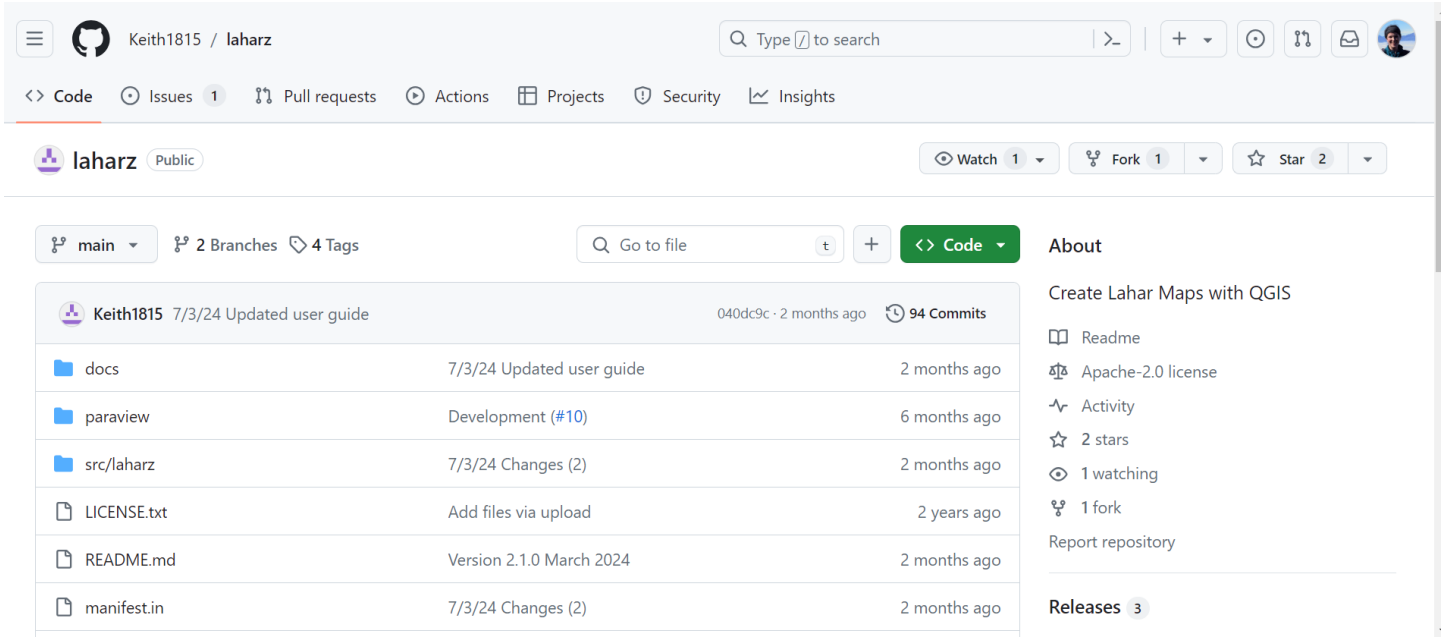


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CSAV 2024 & beyond: LAHARZ has options!

- Keith Blair stand-alone python version
- https://github.com/Keith1815/laharz





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CSAV 2024 & beyond: LAHARZ has optic

LaharZ

University of BRISTOL

Version: 2.1.1 stand alone

- VICTOR platform (using QGIS and python-version of LAHARZ by Keith Blair)
- https://victor.2i2c.cloud/
- Nice because you don't have to worry about installing python dependencies, so we'll use it for class



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USGS

VDAP

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Sinabung, Sumatra, Indonesia
https://www.youtube.com/watch?v=DHNFNH3MODE (surge front and sabo dams)

Japan
http://www.youtube.com/watch?v=kznwnpNTB6k (surge front)
http://www.youtube.com/watch?v=52C5lURB9Pg

Alaska, Pavlof: flat topography and channel confinement
http://www.youtube.com/watch?v=oGLAbsj_C_Y

Philippines
https://www.youtube.com/watch?v=atLvPctJjus

Pyroclastic Flows
Sinabung http://www.youtube.com/watch?v=-iSFM-T-F_0
Unzen http://www.youtube.com/watch?v=Cvjwt9nnwXY
Krakatau http://www.youtube.com/watch?v=JmPuy-pqIQE
Paluweh http://www.youtube.com/watch?v=N4H0KrMPEyw

Mount St Helens, USA: Blast, Landslide/ Avalanche, Lahars, Pyroclastic flows, secondary blasts
http://www.youtube.com/watch?v=rbgAOfv-W20

Vesuvius- Pompei re-creation: https://www.youtube.com/watch?v=dY_3ggKg0Bc